

Task-2

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    // Open source file for reading
    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open destination file for writing
    // Create it if it does not exist
    // Truncate it if it already exists
    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    // Read from source and write to destination
    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead)
        {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }

    if (bytesRead < 0)
    {
        perror("Error reading source file");
    }

    // Close both files
    close(source);
    close(destination);

    printf("File copied successfully.\n");
    return 0;
}
```

Compilation, Input & Output :

```
bodanapurnachandarrao 10947  0.0  0.0 435383280  3680  ??  Ss  10:48AM  0:00.01 /System/Library/Frameworks/NetFS.framework/Ve
bodanapurnachandarrao@boddanas-MacBook-Air Practical-1 % cd ..
bodanapurnachandarrao@boddanas-MacBook-Air Practical % mkdir Practical-2
bodanapurnachandarrao@boddanas-MacBook-Air Practical % cd Practical-2
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % nano Practical-2.c
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % echo "This is the content of the source file." > source.txt
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % cat source.txt
This is the content of the source file.
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % gcc Practical-2.c -o Practical-2
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % ./Practical-2
File copied successfully.
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % █

bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % nano Practical-2.c
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % cat destination.txt
This is the content of the source file.
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % cat source.txt
Hello, this is my OS practical.
bodanapurnachandarrao@boddanas-MacBook-Air Practical-2 % █
```